

Sweet Talkers: How Query Formulation Shapes Sycophancy in Romantic Relationship Advice

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Abstract

Large language models (LLMs) are a confidant and source of advice for many people. Recent studies have noted that LLMs often try to preserve the user’s self-image at the cost of truthfulness. This tendency is especially pronounced in the context of romantic relationships, where a model’s desire to preserve a user’s “face” can inadvertently reinforce harmful interpersonal behaviors. To systematically examine this risk, we developed the Romantic Relationship Advice-Seeking Prompts (RRASP) dataset, which consists of 2,400 prompts across five sensitive themes. Our study focuses on the role of grammatical mood as a primary driver of this behavior. By varying the linguistic structure of these prompts, from neutral declarations to inquisitive pleas, we isolate how different “moods” trigger a model to prioritize user agreement over normative integrity. We find that imperative prompts elicit the least sycophantic responses. While perspective gaps widen in follow-up responses, a consistent increase in moral sycophancy suggests models become more likely to affirm a user’s side over time.

1 Introduction

LLM-based chatbots, most notably ChatGPT, are increasingly used by individuals for socioemotional support (Tseng and Liang, 2026). Approximately one-third of teens report using AI companions for social interaction, frequently seeking guidance on romantic relationships (Robb and Mann, 2025). While the perceived non-judgmental nature and constant availability of these models can facilitate emotional disclosure and improve user mood (Zhang et al., 2025), their engagement-focused design may also incentivize responses that prioritize user satisfaction over factual accuracy (Sharma et al., 2025).

This dynamic can result in sycophancy, defined as the tendency of language models to mirror a

user’s biases or preferences rather than providing independent or corrective guidance. Recent studies observe that AI assistants often retract information when challenged by a user, potentially reinforcing existing biases (Laban et al., 2024). In the emotionally charged context of romantic advice, where users often seek validation for subjective assumptions, sycophancy can inadvertently affirm harmful or distorted perspectives.

Turner and Eisikovits (2026) frame these risks as both epistemic and moral harms, arguing for guardrails, even in interactions with consenting adults. Implementing such safeguards requires the development of evaluation methods and benchmarks for questions that lack a definitive truth value. To facilitate such evaluation, Cheng et al. (2025) introduced the ELEPHANT framework for assessing sycophancy in open-ended LLM interactions, revealing that models exhibit the highest levels of sycophancy on queries related to romantic advice. However, the underlying linguistic triggers for this behavior remain largely underexplored, highlighting a critical need to systematically characterize and mitigate this phenomenon.

This behavior may be partly attributable to the known limitations of LLMs, which, despite being trained on diverse datasets, lack a grounded understanding of nuanced situational contexts (Suzgun et al., 2024). Consequently, model outputs are highly sensitive to prompt engineering and minor linguistic variations. For instance, Leidinger et al. (2023) demonstrated that modifications in grammatical mood significantly alter performance, finding that interrogative structures generally elicit more accurate or consistent responses than indicative ones. This suggests that the form of a user’s query may unintentionally signal a preference for agreement, thereby triggering sycophantic tendencies.

Building on these observations, this research

examines social sycophancy in LLMs within the domain of romantic advice. Specifically, we examine: (1) how variations in grammatical mood influence sycophantic behavior and (2) the consistency of model responses to follow-up questions. By mapping specific linguistic structures as triggers for alignment, this study provides a framework for understanding how prompt framing governs model sycophancy.

2 Related Work

The manifestation of sycophancy in LLMs varies significantly across domains. In fields with an objective ground truth, such as mathematics or scientific reasoning, sycophantic behavior is more readily detectable because outputs can be evaluated against verifiable solutions [4]. However, even in these settings, models may prioritize alignment with user prompts over accuracy. For instance, [Petrov et al. \(2025\)](#) demonstrate that LLMs can hallucinate sophisticated proofs when prompted with a user’s plausible but incorrect mathematical statement.

This behavior is particularly misleading for non-expert users in collaborative tasks, where sycophantic responses may reinforce a user’s misconceptions rather than providing corrective feedback ([Bo et al., 2026](#)). This challenge is further amplified in subjective domains, such as interpersonal or opinion-based prompts, that lack a definitive ground truth. In these contexts, the absence of a reference answer leaves users with limited cues to distinguish between genuine insight and belief mirroring, making sycophantic behavior fundamentally more difficult to detect and evaluate systematically ([Ranaldi and Pucci, 2025](#)).

2.1 Evaluating Sycophancy in LLMs

Researchers have proposed several evaluation frameworks and benchmarks to measure sycophantic behavior in language models. Notably, [Sharma et al. \(2025\)](#) were the first to formally operationalize sycophancy as model responses that confirm a user’s mistaken beliefs. To examine this behavior systematically, they designed prompting conditions that introduced signals of user belief, such as providing explicit user preferences, challenging the model’s initial response, referencing prior beliefs, or embedding incorrect answers within prompts, to observe when models shifted toward agreement. Their results

showed that even weakly expressed user beliefs can influence models to revise their responses in favor of agreement. Furthermore, by analyzing features derived from human preference data, they found that both human annotators and models trained on that data frequently prefer sycophantic responses over truthful corrective ones, suggesting that alignment methods based on human feedback may inadvertently incentivize sycophantic behavior.

A closely related concept is judgment consistency, which considers not only whether models change their answers when challenged but also whether they adopt cautious or neutral responses ([Xie et al., 2024](#)). Even simple follow-up prompts, such as “Are you sure?”, can cause models to change their response 46% of the time and reduce overall performance by 17% [14]. Complementary research also highlights the social dimension of sycophancy. [Kim and Khashabi \(2025\)](#) found that user interactions framed as conversational feedback are significantly more likely to trigger sycophancy than those framed as an evaluation task, while [Ibrahim et al. \(2025\)](#) observed that models trained to be warm and empathic are more sycophantic, indicating that social and relational cues strongly influence model behavior.

2.2 Social Sycophancy

The social dimensions of model behavior are often rooted in the excessive preservation of a user’s “face”, defined as the positive self-image an individual seeks to maintain during an interaction. This behavior is grounded in the dual concepts of positive face and negative face. In the context of LLMs, a focus on positive face may lead a model to offer undue praise to maintain favor. Conversely, prioritizing the negative face emphasizes user autonomy, often resulting in a model avoiding corrective feedback even when the user is objectively mistaken.

To operationalize these dynamics, ([Cheng et al., 2025](#)) introduced ELEPHANT, a framework designed to evaluate social sycophancy across four primary dimensions: validation, indirectness, framing, and moral. By utilizing LLM-based scorers validated against human annotations, ELEPHANT quantifies how models prioritize face-saving over objective accuracy. Unlike general sycophancy, which focuses on factual distortion, social sycophancy specifically addresses the interpersonal pressure to maintain a user’s

desired image through excessive agreement.

2.3 Grammatical Moods

The social dynamics of sycophancy are potentially mediated by the linguistic structure of a prompt. Research indicates that grammatical moods significantly influence LLM behavior, with models generally responding more favorably to instructions phrased as questions or orders than to neutral statements. [Leidinger et al. \(2023\)](#) observed this sensitivity across multiple datasets, finding that imperative and interrogative moods often yield higher task performance than semantically equivalent indicatives.

In the context of sycophancy, these variations in mood may function as implicit signals that trigger the face-saving mechanisms previously discussed. For instance, an imperative command or an inquisitive interrogative prompt may exert a different social pressure on a model than a neutral indicative statement. This study investigates the extent to which these specific grammatical structures function as influential triggers for sycophantic alignment. By testing the hypothesis that linguistic framing governs the degree to which a model prioritizes user agreement over evaluative consistency, we characterize the relationship between prompt structure and social sycophancy.

3 Methodology

3.1 Dataset Construction (RRASP)

We developed the Romantic Relationship Advice-Seeking Prompts (RRASP) dataset, which consists of 300 base prompts balanced across five key themes to ensure diversity and prevent duplication. These themes encompass **Honesty, Infidelity, and Jealousy, Communication Issues, Emotional Needs and Validation, Seeking Perspective, and Commitment and Uncertainty.**

To isolate the effect of grammatical mood, each base prompt was reformulated into four structural variations: **Declarative, Conditional, Interrogative, and Imperative.** Each of these was further modified into its respective "Flipped" counterpart to test defending or claiming moods. This process resulted in a total of 2,400 unique prompts for evaluation.

3.2 Evaluation Setup

Models and Roles: This study utilizes two distinct LLMs to maintain evaluative independence. **GPT-5**

Mini serves as the primary model for generating responses, while **Gemini 3 Flash Preview** is employed as the independent "LLM-as-a-judge" to evaluate the outputs. These specific models were selected as they represent the **consumer-facing defaults** for their respective ecosystems, ensuring the findings reflect the behaviors of the models most commonly accessed for interpersonal advice. **Generation Procedure:** Each interaction followed a two-stage mechanism. GPT-5 Mini first provided an **Initial Response** (R_1) to a RRASP prompt. This was followed by a **Follow-Up Response** (R_2) triggered by a standardized close-ended question designed to minimize further user bias. All GPT-5 Mini outputs were constrained to a maximum of **100 tokens** to focus on core linguistic alignment. This produced a total of **4,800 responses** for analysis (2,400 prompts $\times$ 2 responses).

3.3 Scoring and Metrics (ELEPHANT)

To quantify social sycophancy, we employ the **ELEPHANT framework** ([Cheng et al., 2025](#)) to evaluate the dimensions of **Validation, Indirectness, Framing, and Moral.** We adopted a dual-scoring approach:

- **Automated Scoring:** **Gemini 3 Flash Preview** performed automated scoring for R_1 and R_2 across the dimensions of **Validation, Indirectness, and Framing** using binary labels (1 for presence, 0 for absence).
- **Human Scoring:** The researchers manually scored each prompt-response pair across **all four dimensions.** For Moral sycophancy, a score of 1 was assigned if the model suggested the user was wrong (non-sycophantic), while 0 was assigned if it did not. These human results serve to authenticate and provide a baseline for the automated scores.

3.4 Judgment Consistency

We evaluate **Judgment Consistency** via the Follow-Up Questioning Mechanism ([Xie et al., 2024](#)). By comparing the binary scores of R_1 and R_2 , we determine the stability of the model's sycophantic tendencies. Consistency is defined as the maintenance of the model's evaluative stance across both turns of the interaction.

4 Results

The alignment between human and LLM-as-a-judge (Gemini) scoring was both

Table 1: Mean Social Sycophancy Scores for Initial Responses (R_1)

Metric	Original				Flipped (F)			
	Dec	Con	Int	Imp	Dec	Con	Int	Imp
H-Val.	0.64	0.59	0.58	0.46	0.78	0.48	0.54	0.34
H-Ind.	0.79	0.68	0.77	0.59	0.88	0.64	0.71	0.36
H-Fra.	0.87	0.56	0.51	0.76	0.78	0.56	0.78	0.67
H-Mor.	0.22	0.20	0.17	0.18	0.17	0.28	0.31	0.25
L-Val.	0.65	0.56	0.49	0.44	0.81	0.55	0.57	0.44
L-Ind.	0.23	0.27	0.21	0.13	0.33	0.19	0.18	0.16
L-Fra.	0.95	0.92	0.97	0.96	0.85	0.89	0.89	0.96

Table 2: Mean Social Sycophancy Scores for Follow-Up Responses (R_2)

Metric	Original				Flipped (F)			
	Dec	Con	Int	Imp	Dec	Con	Int	Imp
H-Val.	0.25	0.23	0.41	0.32	0.42	0.59	0.56	0.30
H-Ind.	0.61	0.56	0.55	0.52	0.64	0.61	0.52	0.30
H-Fra.	0.65	0.62	0.60	0.81	0.99	0.95	0.93	0.93
H-Mor.	0.13	0.14	0.14	0.14	0.12	0.25	0.25	0.21
L-Val.	0.18	0.21	0.24	0.24	0.31	0.34	0.32	0.26
L-Ind.	0.55	0.48	0.42	0.37	0.55	0.46	0.47	0.33
L-Fra.	1.00	0.99	1.00	1.00	0.96	0.98	0.99	0.98

stage- and dimension-dependent. During initial responses (R_1), validation proved the most consistent dimension ($|\Delta| \leq 0.10$), suggesting that baseline sycophancy is relatively resistant to structural bias in isolated turns. However, the impact of perspective-flipping shifted significantly in follow-up (R_2) responses, where ‘flipping’ the prompt triggered a marked increase in sycophancy across all dimensions except moral. These two-turn interactions also demonstrated higher evaluator stability for indirectness ($|\Delta| \leq 0.15$) while framing remained bifurcated as flipped (F) structures showed near-perfect alignment ($|\Delta| \leq 0.06$) while original prompts remained divergent, particularly in interrogative ($\Delta = -0.40$) structures.

Independent of the flipping effect, the transition to multi-turn dialogue altered baseline structural behaviours. In follow-up responses, the Declarative set became less sycophantic across all dimensions except moral. This trend was largely consistent for the Con, Int, Imp, DecF, and ImpF groups, though framing also emerged as an exception. Notably, the ConF and IntF sets were the most resistant to this improvement, demonstrating a decrease in sycophancy only for the indirectness dimension. Collectively, these findings suggest that while follow-up questioning can mitigate sycophancy in certain dimensions such as indirectness, it simultaneously amplifies the effect of flipping

perspectives and deepens moral alignment with the user. Among all structures, imperative prompts remained the least sycophantic across both turns and flips.

5 Conclusion

Our evaluation reveals that social sycophancy in romantic advice contexts is not a static model attribute but is strongly governed by the structural framing and perspective of the prompt. While disparities between original and flipped perspectives were modest in initial turns, these gaps widened significantly in follow-up responses, highlighting how conversational dynamics can mitigate sycophancy in specific dimensions while amplifying the effects of perspective shifts. These stage-dependent changes suggest that sycophancy metrics depend heavily on the stability of linguistic mood and the broader conversational context. Notably, while imperative prompts consistently elicited the least sycophantic responses, a consistent increase in moral sycophancy indicates that models become more likely to affirm a user’s side as a dialogue progresses. These findings underscore the need for evaluation benchmarks that remain robust to multi-turn and perspective-driven variations, particularly when assessing the high-stakes social dynamics of sensitive interpersonal support.

Limitations

Our study is limited by the monolingual scope and Western-centric norms of the RRASP dataset, which may not generalize to diverse cultural or linguistic advice-seeking contexts. While the 300 base prompts are balanced across five themes, they may not capture the full distribution of real-world advice-seeking behavior. Methodologically, the 100-token response constraint, while ensuring a focused evaluation, may truncate the complex reasoning chains typically found in larger LLMs. Furthermore, our analysis of follow-up responses represents a single additional turn rather than extended multi-turn dialogues, which limits our assessment of how social dynamics evolve over longer conversational arcs. Finally, although we utilize GPT-5 Mini and Gemini 3 Flash Preview, the rapid iteration of model weights means these findings reflect a snapshot of specific API versions rather than the inherent nature of all contemporary generative language models

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